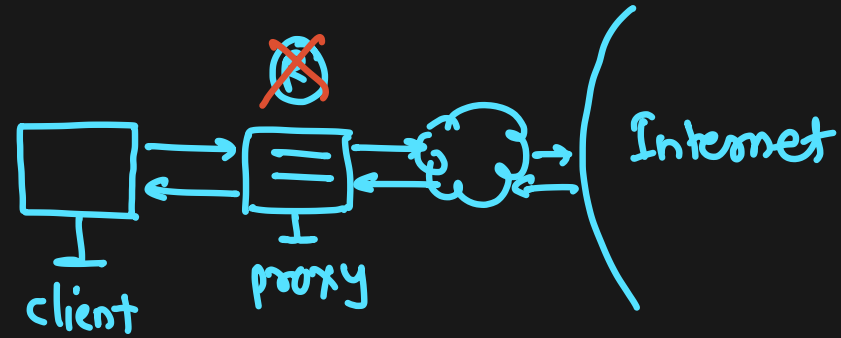




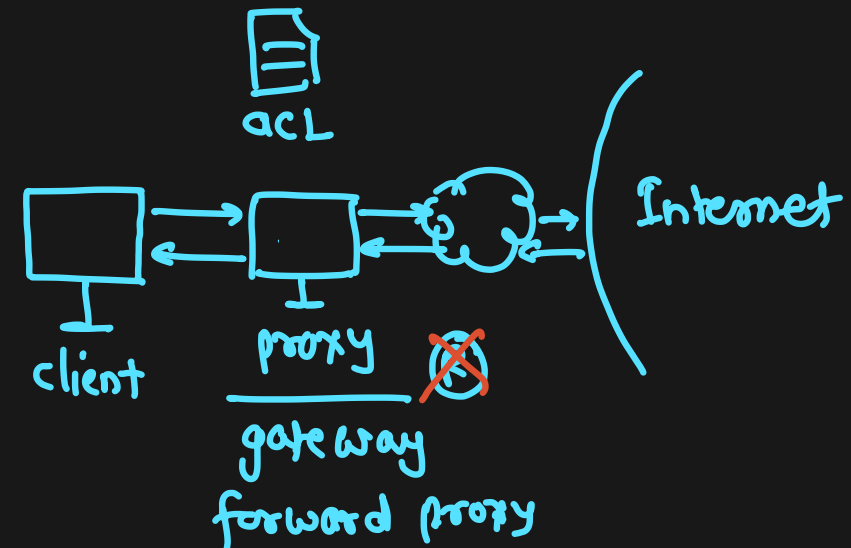
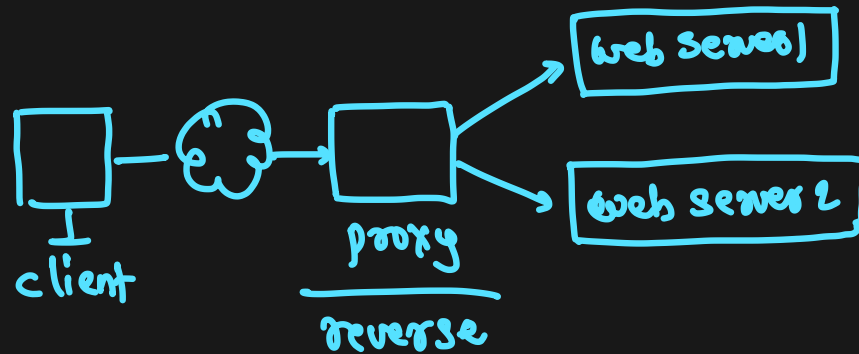
# Proxy Server

Squid Proxy

# Proxy Server



- A proxy server is an intermediary application or system that acts as a gateway between a client (user's device) and a destination server (website, service, etc.)
- Instead of connecting directly to the target server, the client connects to the proxy, which then forwards the request on behalf of the client
- **Key terminologies**
  - Forward Proxy: Sits between internal clients and the external internet → squid proxy
  - Reverse Proxy: Sits between external clients and internal web servers → nginx, httpd
  - Upstream Server: The destination server the proxy connects to
  - Origin Server: The original server hosting the content



# Proxy server flow



## ▪ Client Configuration

3128

- User configures their browser/OS to use a proxy (IP address + port number)

## ▪ Request Forwarding

- Client sends request to proxy instead of directly to the destination

## ▪ Proxy Processing

↪ ACL

- Proxy evaluates the request based on rules (allow/deny, caching, authentication)

## ▪ Request to Server

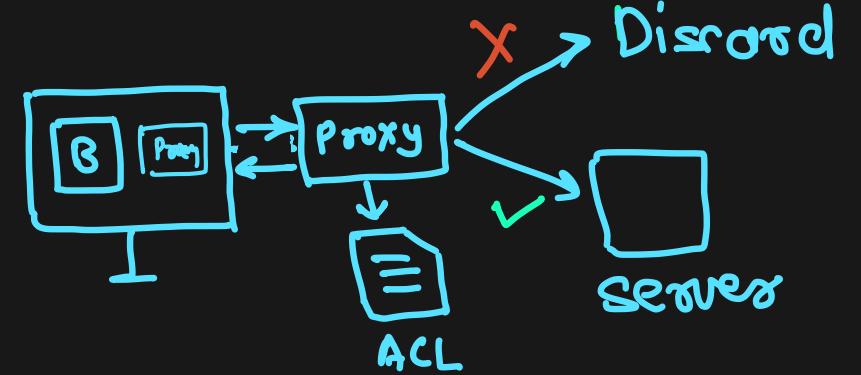
- If allowed, proxy forwards request to the destination server (using its own IP address)

## ▪ Response Received

- Destination server sends response back to the proxy

## ▪ Response to Client

- Proxy forwards the response to the original client



# Functions and use cases



## ■ Access Control & Filtering

- Block specific websites (social media, adult content, streaming)
- Restrict internet access during work hours
- Whitelist only approved domains
- Prevent data exfiltration

## ■ Anonymity & Privacy

- Hide client's real IP address
- Bypass geographic restrictions (geo-blocking)
- Prevent websites from tracking user location
- Avoid IP-based rate limiting

## ■ Performance Improvement (Caching)

- Store frequently accessed content locally
- Reduce bandwidth consumption (by 30-50% in some environments)
- Speed up page load times for repeated requests
- Offload origin servers

## ■ Security

- Act as a barrier between internal network and internet
- Scan incoming/outgoing traffic for malware
- Prevent direct access to internal servers
- Terminate SSL/TLS connections for inspection

## ■ Load Balancing (Reverse Proxy)

- Distribute incoming traffic across multiple backend servers
- Handle failover if a server goes down
- Perform health checks on backend servers

## ■ Bypassing Censorship

- Circumvent government or ISP-level blocking
- Access region-restricted content (Netflix, BBC iPlayer, etc.)

# Types



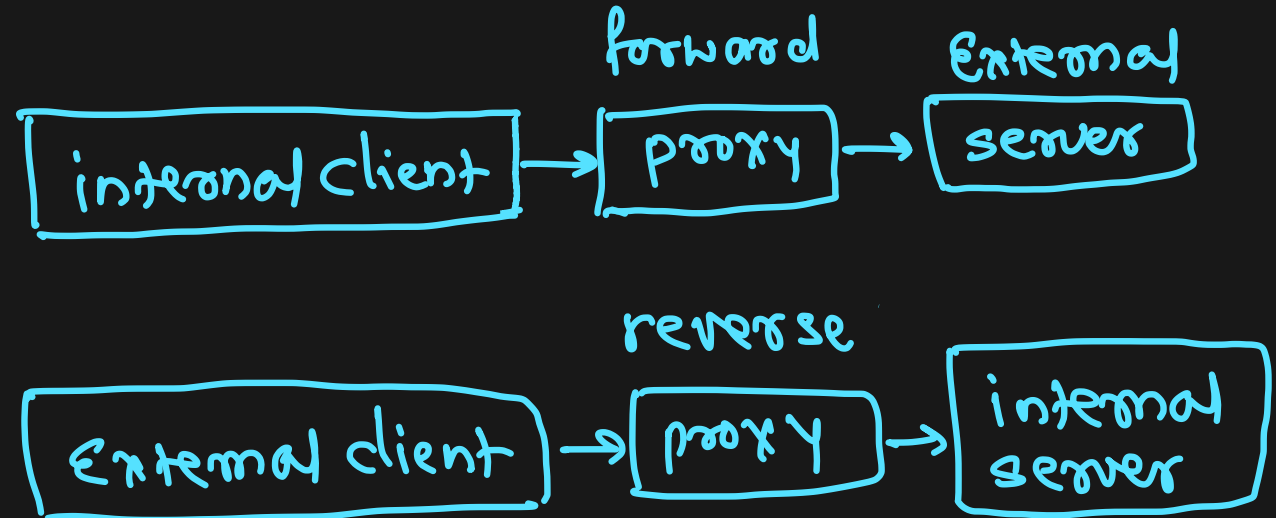
## ■ By Direction / Placement

### ■ Forward Proxy

- Client - Proxy - Internet
- Access control, caching, anonymity
- Internal users

### ■ Reverse Proxy

- Internet - Proxy - Web Server
- Load balancing, security, SSL termination
- External users



## ■ By Protocol Support

| Proxy Type          | Protocol           | Typical Port      | Features                                           |
|---------------------|--------------------|-------------------|----------------------------------------------------|
| ✓ <u>HTTP Proxy</u> | <u>HTTP/HTTPS</u>  | <u>3128, 8080</u> | <u>Web traffic only</u> → squid                    |
| <u>SOCKS4 Proxy</u> | <u>Any TCP</u>     | <u>1080</u>       | <u>No authentication, no DNS</u>                   |
| <u>SOCKS5 Proxy</u> | <u>Any TCP/UDP</u> | <u>1080</u>       | <u>Authentication, DNS resolution, UDP support</u> |
| <u>FTP Proxy</u>    | <u>FTP</u>         | <u>21</u>         | <u>File transfer protocol only</u>                 |

# Popular proxy servers



| Software                              | Type                                 | Platform              | Key Features                                      |
|---------------------------------------|--------------------------------------|-----------------------|---------------------------------------------------|
| ✓ <u>Squid</u>                        | <u>Forward + Reverse + Caching</u>   | <u>Linux/Unix</u>     | <u>High performance, ACLs, SSL bump</u>           |
| ✓ <u>HAProxy</u>                      | <u>Reverse Proxy + Load Balancer</u> | <u>Linux/Unix</u>     | <u>TCP/HTTP load balancing, high availability</u> |
| ✓ <u>Nginx</u>                        | <u>Reverse Proxy + Web Server</u>    | <u>Cross-platform</u> | <u>Lightweight, high concurrency</u>              |
| ✓ <u>Apache (mod_proxy)</u>           | <u>Reverse + Forward</u>             | <u>Cross-platform</u> | <u>Integrated with Apache web server</u>          |
| <u>Privoxy</u>                        | Forward (non-caching)                | Cross-platform        | Ad blocking, privacy filtering                    |
| <u>Tinyproxy</u>                      | Lightweight Forward                  | Linux                 | Minimal resource usage                            |
| <u>SOCKS5 Proxies (Dante, 3proxy)</u> | SOCKS protocol                       | Linux/Unix            | Generic TCP/UDP forwarding                        |



# Squid



# Squid



- Squid is a powerful, open-source caching proxy server for HTTP, HTTPS, and FTP
- It acts as an intermediary between clients and servers, storing frequently requested content to improve speed and reduce bandwidth usage → caching  
↳ forward
- **Primary Use Cases**
  - ✓ **Forward Proxy (Cache):** Internal clients access the external web. Squid caches web pages so subsequent requests are served instantly without re-downloading
  - **Reverse Proxy:** Placed in front of web servers to cache outgoing content, reduce load on the backend, and speed up delivery to external users
- **Key Advantages**
  - **Bandwidth Optimization:** Redundant downloads are eliminated → caching
  - **Speed Improvement:** Cached objects load instantly
  - **Traffic Filtering:** Restrict access to specific websites → ACL
  - **Anonymity:** Can hide client IP addresses

# Installation

## ■ Installation

- sudo yum update
- sudo yum install squid -y

## ■ Post installation verification

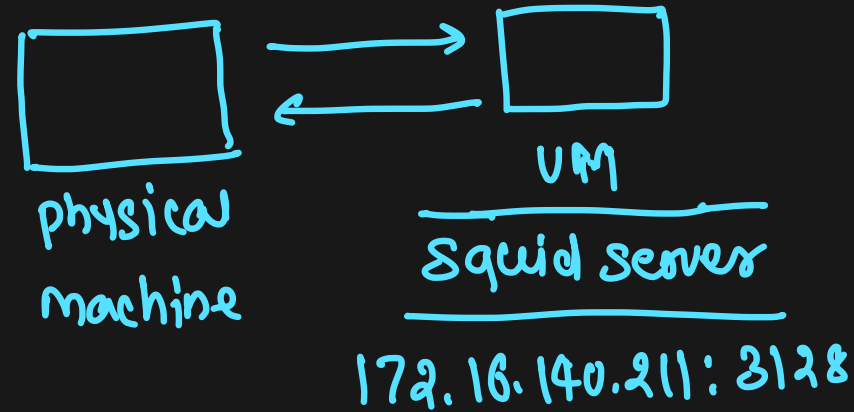
- squid -v # check the version
- squid -k parse # validate the configuration syntax

## ■ System commands

- sudo systemctl start squid # Start the service
- sudo systemctl stop squid # Stop the service
- sudo systemctl restart squid # Restart (required after config changes)
- sudo systemctl status squid # Check if running
- sudo systemctl enable squid # Start on boot

## ■ Configuration file

- The main configuration file is: /etc/squid/squid.conf



# Directives



| Directive                  | Purpose                                                          |
|----------------------------|------------------------------------------------------------------|
| <u>http_port</u>           | <u>Sets the listening port for the proxy.</u> — 3128             |
| ✓ <u>http_access</u>       | <u>Defines who can use the proxy. Order matters!</u>             |
| <u>cache_mem</u>           | <u>RAM used for in-memory cache (Squid uses more than this).</u> |
| <u>cache_dir</u>           | <u>Disk storage location and size (in MB).</u>                   |
| <u>visible_hostname</u>    | <u>Hostname shown in error messages.</u>                         |
| <u>maximum_object_size</u> | <u>Largest cached object size.</u>                               |



# Access Control List (ACL)

- ACLs define who can connect and what they can do
- Rules are processed top-down

```
# Define the ACL
acl localnet src 192.168.1.0/24
# Apply the rule (Allow first, then deny all)
http_access allow localnet
http_access deny all
```

*Handwritten annotations:*

- A red arrow points from the word "source" to the "src" keyword in the ACL definition.
- The word "temp name" is written below "localnet" and underlined.
- The words "ip address" are written below "192.168.1.0/24" and underlined.
- The words "http\_access allow localnet" are underlined.
- The words "http\_access deny all" are underlined.

# Blocking websites



## ■ Method A: Block Individual Domains

```
# In /etc/squid/squid.conf  
acl blocked_sites dstdomain facebook.com youtube.com twitter.com  
http_access deny blocked_sites
```

## ■ Method B: Block from External File

```
# In /etc/squid/squid.conf  
acl blocked_sites dstdomain "/etc/squid/blocked_domains.txt"  
http_access deny blocked_sites
```

### ■ External file /etc/squid/blocked\_domains.txt

```
.facebook.com  
.youtube.com  
.twitter.com  
.instagram.com  
.tiktok.com
```

# Blocking File Downloads by Extension



## ▪ Method A: Block by URL Path Extension

```
# In /etc/squid/squid.conf  
acl blocked_files urlpath_regex -i \.exe$ \.zip$ \.mp3$ \.avi$ \.mp4$  
http_access deny blocked_files
```

## ▪ Method B: Block from External File

```
acl blocked_files urlpath_regex -i "/etc/squid/blocked_extensions.txt"  
http_access deny blocked_files
```

### ▪ External file /etc/squid/blocked\_extensions.txt:

```
\.exe$  
\.zip$  
\.rar$  
\.mp3$  
\.mp4$  
\.avi$  
\.mkv$
```

# Time-Based Access Control



## ▪ Block Social Media During Work Hours

```
# Define work hours (Monday to Friday, 9 AM to 5 PM)
```

```
acl workhours time MTWHF 09:00-17:00
```

```
# Define social media sites
```

*"/etc/squid/urlstat"*

```
acl social_media dstdomain facebook.com twitter.com instagram.com
```

```
# Block social media only during work hours
```

```
http_access deny social_media workhours
```

```
# Allow outside work hours (will be caught by later allow rules)
```

```
http_access allow all
```

# Break Time Schedule



- Day abbreviations

- M=Monday, T=Tuesday, W=Wednesday, H=Thursday, F=Friday, A=Saturday, S=Sunday

```
# Allow certain sites only during break time
acl break_time time MTWHF 12:00-13:00
acl allowed_during_break dstdomain news.com reddit.com

http_access allow allowed_during_break break_time
http_access deny allowed_during_break # Block at other times
```

# Blocking Ads and Banners



## ■ Basic Banner Blocking

```
# Define ad-related URL patterns
acl ad_urls url_regex -i \
    "/etc/squid/ad_patterns.txt"

# Redirect to a transparent GIF instead of showing error
deny_info NOTE_ADS_FILTERED ad_urls

# Deny access to ad URLs
http_access deny ad_urls
```

## ■ External file /etc/squid/ad\_patterns.txt

```
/ads/
/advert/
/banner
/doubleclick
^http://ad\.
^http://ads\.
```